

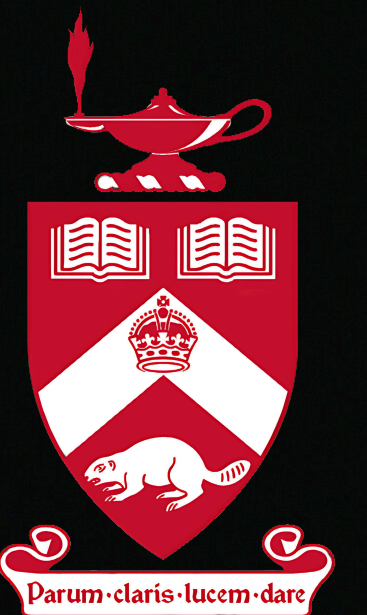


The “High” Mind: Oneirogenic Psychedelics Facilitates Insight Problem-Solving, Predictive Processing & Relevance Realization

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Introduction

Recent studies in cognitive science, psychology, and neuropharmacology have demonstrated that dreams are highly related to psychedelic experiences. In addition, many Nobel prize laureates have experienced moments of insight either through psychedelics or dreams, or both. Despite this, there remains no integrated understanding of the implications that psychedelics may have for insight problem-solving, predictive processing (PP), and relevance realization (RR). In hopes of illustrating the profound utility of psychedelics as an explorative tool, I examined how contemporaneous empirical studies in the literature converge towards this end and propose that psychedelics offer a powerful window into the processes of RR, PP, and cognition as a whole.

Predictive Processing

The **Predictive Processing** (PP) framework, also known as *predictive coding* in neuroscience, is a theory that posits the human mind as a predictor of reality, constantly anticipating its future states whilst using errors to fine-tune its predictions. Dreams are **offline states** that enable the mind to **integrate all prior sensory and semantic information** via **replays** to improve its paradigm of prediction and learning. Given their similarity to dreams, psychedelics may act to do the **same**.

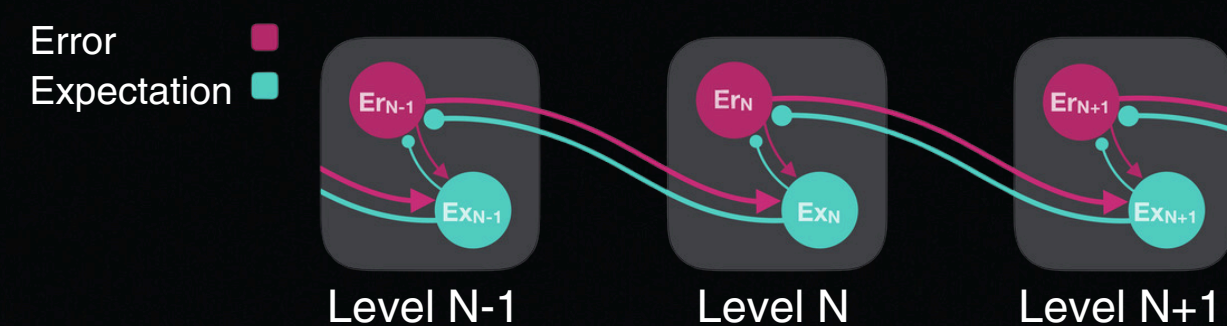


Fig. 1a. A simplified hierarchical PP model based on the visual cortex circuit (Rao & Ballard, 1999; Walsh et al., 2020).

Psychedelics, Dreams & Insight

Insight problem-solving are instances in which previously unsolvable problems are suddenly solved with ease. It has been argued that insight might be a highly efficient form of predictive processing in which the agent engages in what is called “**fact-free learning**” and **meta-cognitive timing prediction errors**. Historically, both dreams and psychedelics have had some role to play in some of the world’s greatest scientific discoveries (e.g., chaos theory, loop quantum gravity theory, coordination theory). **Dreams and psychedelics are powerful catalysts of creativity.**

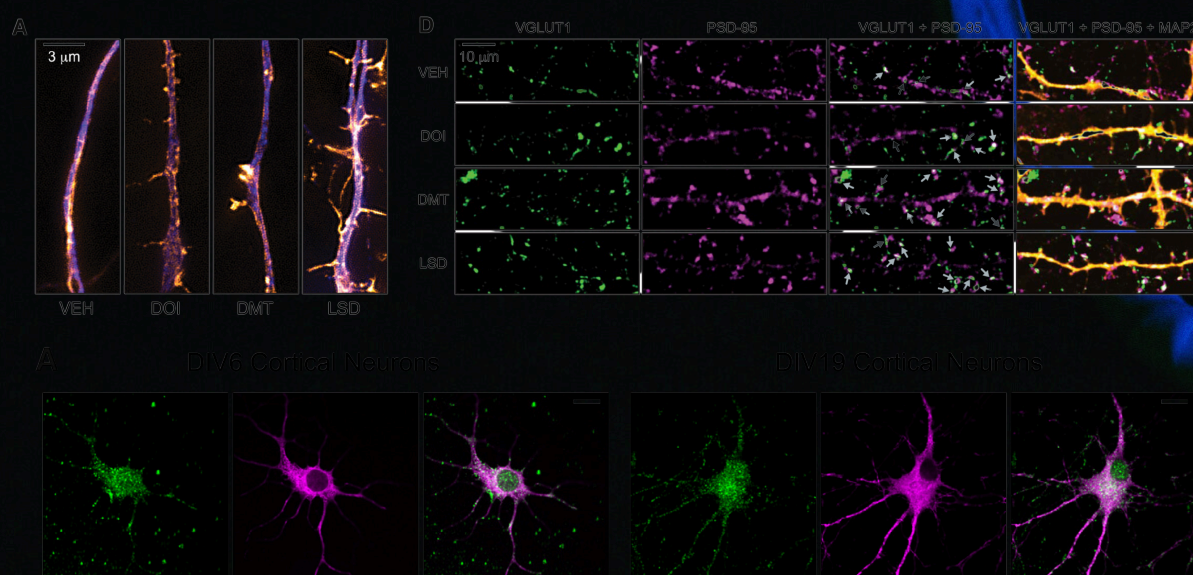


Fig. 1b. Some studies show that psychedelics can even contribute to synaptogenesis, spinogenesis, and functional neuroplasticity (Ly et al., 2018).

Hypotheses & Discussion

Thesis: Psychedelics may offer us insight into the nature of human cognition through its modulations of insight problem-solving, predictive processing, and relevance realization. Key questions addressed include:

- What are the primary function of dream states?
- Why are dreams so similar to psychedelic experiences?
- Can we leverage psychedelic substances for science?
- How do psychedelics impact the brain’s predictive coding?
- How do psychedelics and dreams elicit “Aha!” moments?

Overfitted Brain Hypothesis

The Overfitted Brain Hypothesis (OBH) claims that dreams intentionally simulate virtual experiences that are rife with random noise which is injected into the hierarchical structure of the brain to prevent **overfitting**. Overfitting is a prominent challenge for deep neural networks which must trade-off between selection and generalization (this hypothesis is drawn from experience with ANNs). The often cognitively bizarre nature of dreams is **not random or purposeless** but rather **enables the mind to better generalize and prevent overfitting**. If this is indeed the case, then the corrupted sensory inputs observed in psychedelic experiences may also be simulating the stochastic nature of dreaming.

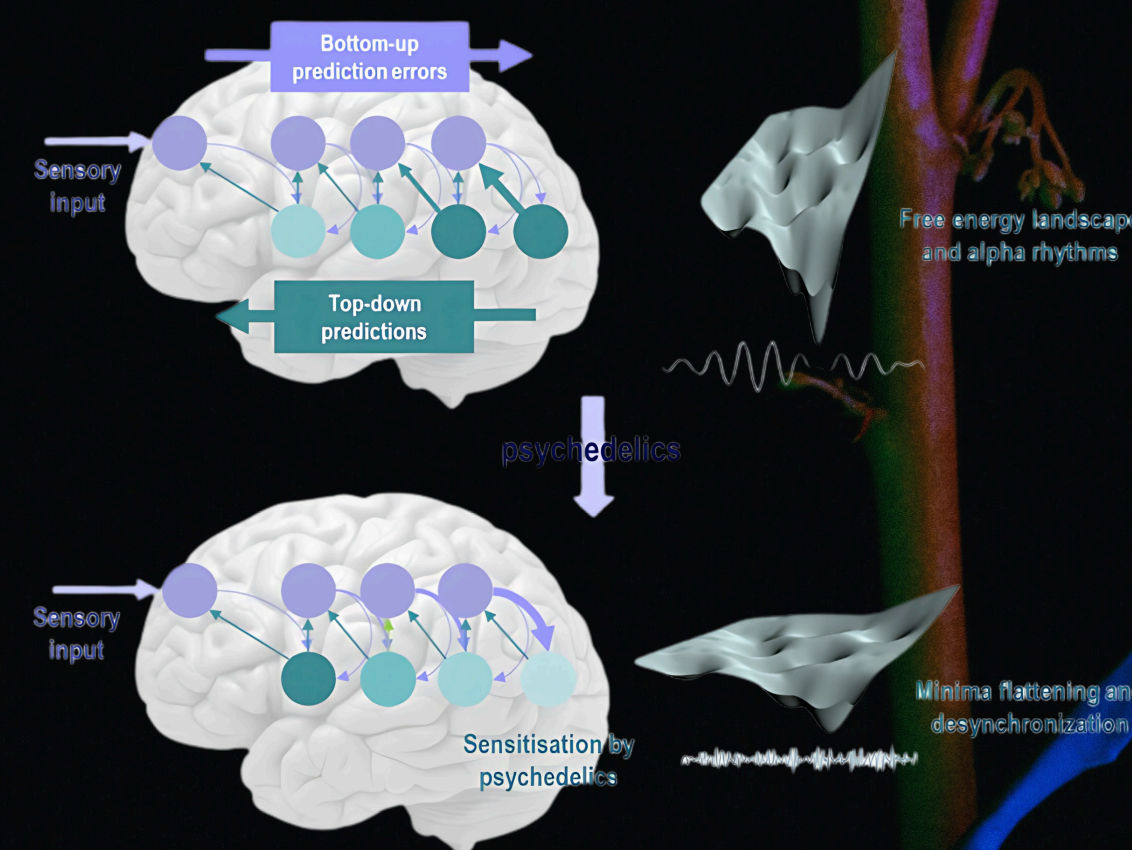


Fig. 2. Proposed effects of psychedelics on hierarchical predictive coding (Carhart-Harris & Friston, 2019).

Relevance Realization

The Relevance Realization (RR) framework is an emerging overarching proposal in cognitive science that asserts that the mind is a dynamical system of hierarchical predictive processing and that most if not all cognitive processes involve identifying **salient** information. The mind **prespecifies** the relevance of information by using a feedback cycle (“sensorimotor loop”) and a set of **enabling** and **selective** constraints. These constraints are **dynamical coupled to each other and are situated within a system of opponent-processing** and serve to “tighten” or “loosen” the fit of information. Such an intricate system ensures an **optimal regulation of error**. This ability to identify relevance may be **elevated and pronounced in psychedelic experiences**.

Summary & Conclusion

- Psychedelics **enrich** the brain’s predictive coding system by **enhancing** the sensorium of generative models.
- Computational neuroscience highlights that psychedelics mirror dream states, which serve as the brain’s natural response to counter problems of **overfitting**.
- Psychedelics **consolidate existing views of RR** as an exaptive and all-encompassing account of cognition. Impacts on the sensorimotor loop enable the realization of relevance to a greater extent than in default consciousness.
- Such integrated understanding **demand more empirical attention** that psychedelics be used more extensively as an explorative tool in cognitive science.
- **Massive** implications for several areas including but not limited to psychopathology, cognitive neuroscience (consciousness), AGI, robotics, and computational modelling of disease and brain states.
- Interestingly, Prince & Richards (2021) have suggested that psychedelics may be treated as **dream substitutions** to promote more **robust learning** or to alleviate the repercussions of sleep deprivation. This underscores the immense utility of psychedelics in a variety of areas.

The REBUS Model

The Relaxed Beliefs under Psychedelics (REBUS) by Carhart-Harris & Friston (2019) has been an influential formulation model of the brain action of psychedelics. It proposes that psychedelics work to relax the precision of high-level priors or beliefs, thereby “liberating” the flow of bottom-up inputs. This causes an increased sensitization of high-level priors to bottom-up signalling, leading to a heightened sensitivity that enables the potential revision and de-weighting of overweighted priors. This may explain key outcomes of psychedelic use such as more efficacy in mental health interventions, instances of insight problem-solving.

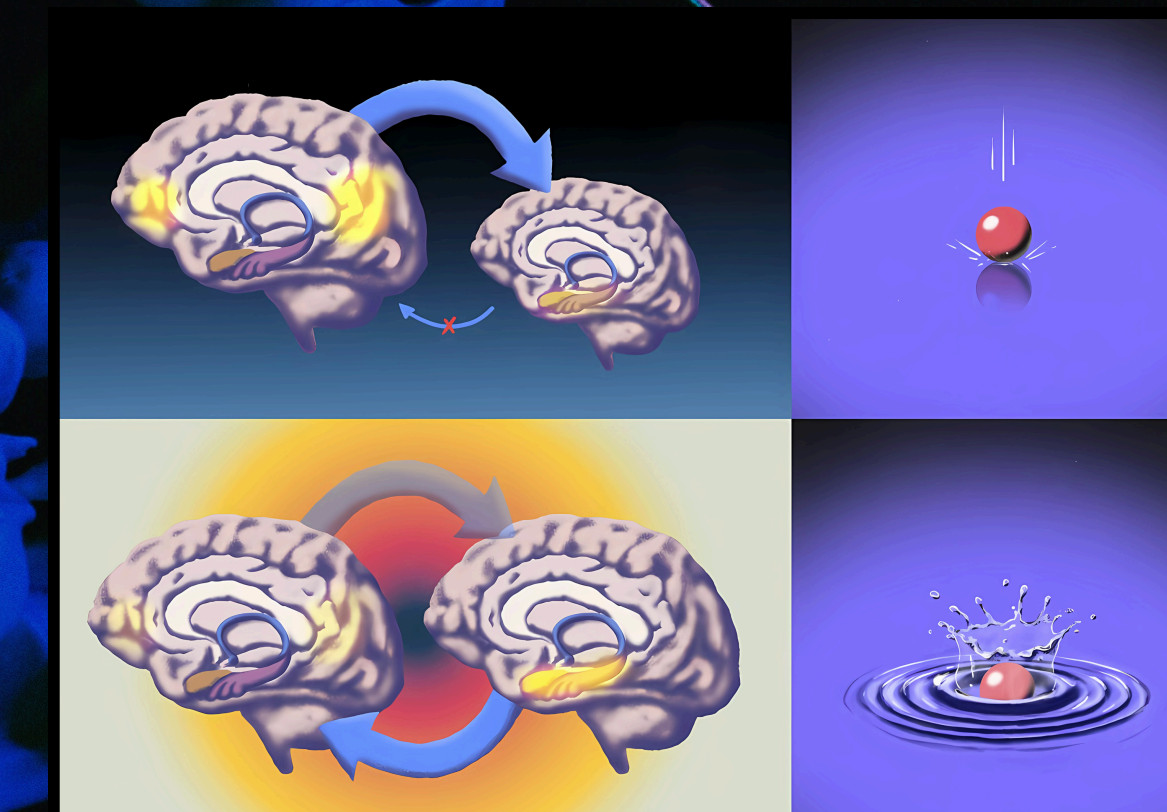


Fig. 3. **Top row:** brain organization in psychopathologies in which high-level priors are overweighted (insensitivity to bottom-up signalling). **Bottom-row:** brain organization under psychedelics in which there is critical slowing, i.e., systems at criticality display maximal sensitivity to perturbation (Carhart-Harris & Friston, 2019).

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